

What Can You Read? What Might Be Wrong?

Module 2 · Day 3, Session 2: Reading the Figures That Carry the Claim

3ie

1 • Why the picture matters

Where we are

- Day 1** is the number answering your question, and is it big enough?
- Day 2** which method produced it, and what does that method assume?
- This morning** what is it worth, and which assumption decides that?
- Now** the chart a decision-maker actually sees

Every session this week has ended with a number and the assumptions behind it. A decision-maker usually sees neither directly: they see a chart and a sentence about it.

An assumption that is left out of the chart is hard to question.

This session is about **reading charts as evidence**. When a report reaches you, do you question its charts the way you have questioned its numbers this week?

Six pairs, two questions

Each section of this session shows the same data drawn two ways. One version is fair; the other invites a wrong reading, or cannot be read correctly at all.

For every pair, in groups:

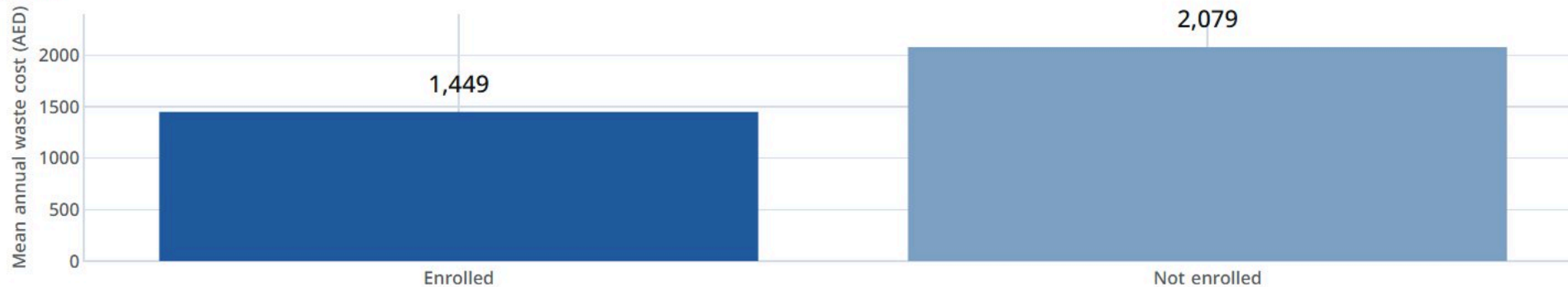
1. What is this figure telling you?
2. What might be wrong or misleading?

The same data, two pictures

Among the businesses that were offered GreenWaste, the ones that enrolled paid **1,449 AED** in waste costs at baseline and the ones that did not paid **2,079 AED**. A gap of **630 AED**.

Before the programme started, the businesses that enrolled were already paying less. Keep this 630 AED; it returns in section 4.

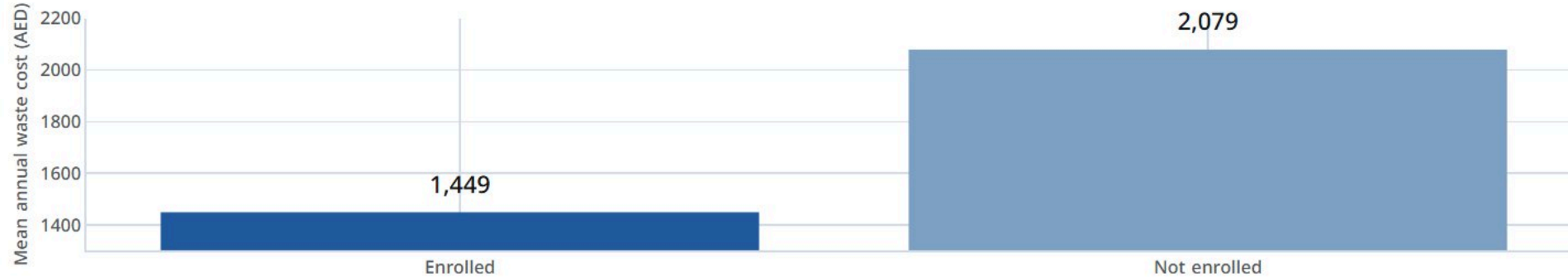
Figure A



Is this figure fair? What would you check first?

The same gap, drawn to fill the frame

Figure B



Same numbers and labels, different impression.

The gap is 630 AED in both figures. Here the y-axis starts at 1,300, so a 30% difference fills most of the frame.

In groups, two minutes. Which figure would you put in a briefing to a minister? Which would you put in an annex? Is your answer the same for both?

[MENTI PLACEHOLDER] Which figure for the minister?

Briefing a minister: **figure A, figure B, or either with one sentence added?**

A · B · Either, with one sentence added

Figure A starts the axis at zero; figure B starts it at 1,300. Go to **menti.com** and enter code **1234 5678**, or scan the code on the right.



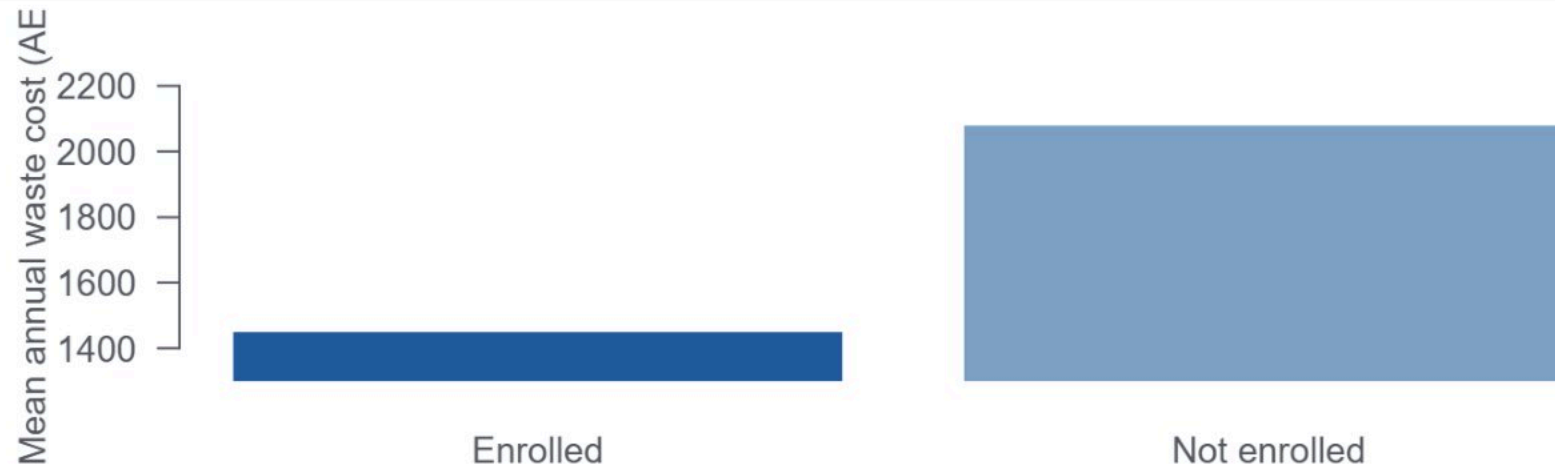
Try it: where should the axis start?

R Code

↻ Start Over

▶ Run Code

```
1 start <- 1300 # where the y-axis begins: try 0, then 1400
2 clean_par()
3 barplot(costs, ylim = c(start, 2300), xpd = FALSE, border = NA,
4         col = c(ACCENT, ACCENT2), ylab = "Mean annual waste cost (AED)")
```



Predict before each run: how big does the gap look from 0? From 1,400?

Rule 1: read the axis before the shape

A truncated axis works like a **magnifying glass**.
That is useful when the question is whether a difference exists at all.

It misleads when the reader does not notice that the axis has been cut.

Test: **cover the bars, read the y-axis, uncover.**

Does the gap still look the same size?

Here the number and the picture agree: 1.13 SD is a large gap. **If the gap really is large, what is still wrong with figure B?**

The gap

630 AED · the same in both figures

As a share

30% of the non-enrolled mean

In SD units

1.13 standard deviations

The accurate line

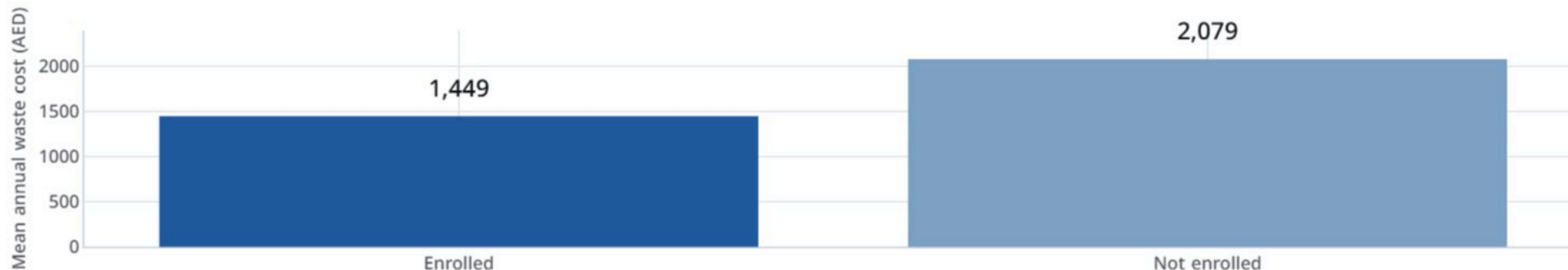
"a substantial gap on a truncated axis"

2 · What the bar is hiding

What the bar of means leaves out

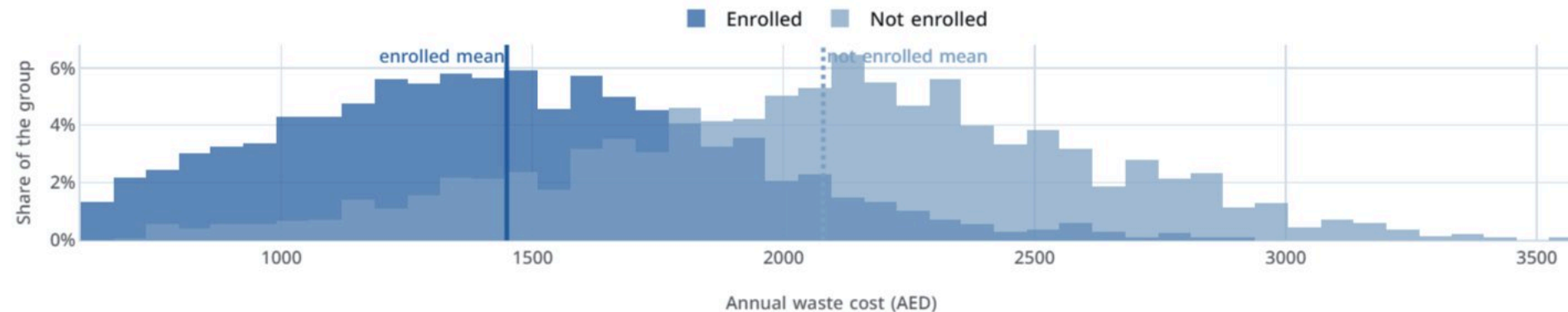
The same comparison as a **bar chart of means**, one number per group. It is one of the most common figures in evaluation reports.

Two groups with the same mean can have little else in common.



Before the next slide: what share of the enrolled businesses do you think paid **more** than the average non-enrolled business? Write a number down.

The same comparison, as distributions



The two groups overlap very little. Only 7.8% of enrolled businesses (231) paid more than the average non-enrolled business. A bar chart of means cannot show this.

Look back at the number you wrote down. Did you expect 7.8%?

Rule 2: ask what the bar is averaging over

A mean	a summary of a group · not a description of anyone in it
Two groups can	share a mean and share nothing else
A bar chart	cannot show how much two groups overlap
So ask	"how much do the two groups overlap?" · it is answerable

This is the same instinct as Day 1's confidence interval and Day 2's baseline balance table. **Ask what the summary leaves out.**

Here the overlap agrees with the gap. What would you conclude if the two groups had overlapped heavily?

3 · Check the denominator

The change nobody can compute

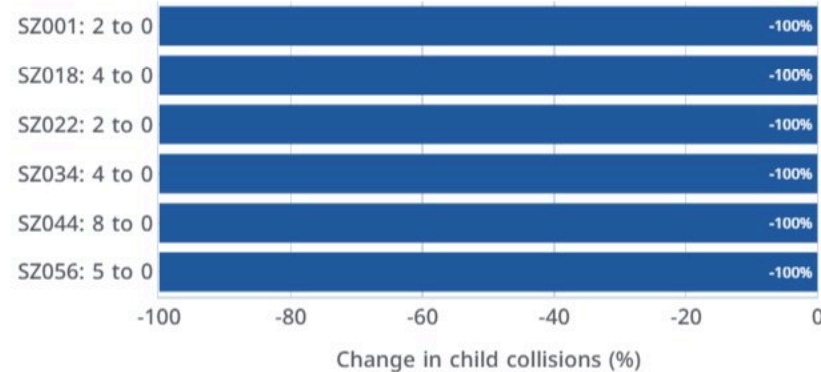
The school-zone trial put its safety package on 200 of its 400 segments and counted child collisions before and after. An analyst wants to report **the percentage change in child collisions at the typical package segment**.

Baseline child collisions	Share of segments
0	5.0%
1	10.5%
2	12.5%
3 to 5	45.5%
6 to 9	22.5%
10 or more	4.0%

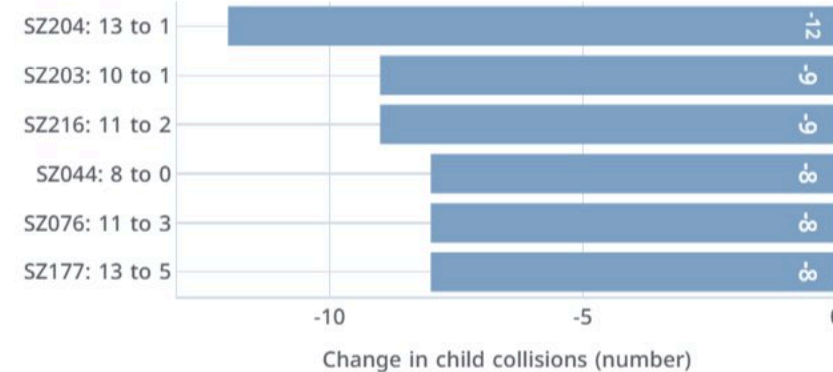
10 of the 200 package segments had no child collisions at baseline, so their percentage change would mean dividing by zero. Any figure that reports one has dropped these segments or handled them somehow. Does it say which?

Two charts, two different “biggest wins”

The largest percentage reductions



The largest absolute reductions



These two lists have one segment in common. Every segment on the left fell to zero, a 100% cut. So did 33 package segments in all, 23 of them from four collisions or fewer. The segments on the right removed 54 between them.

Both are correct. The left answers *where did the package work best?*, the right, *where did it prevent the most harm?* **Which question would a minister want a report's chart to answer?**

Try it: one change, two averages

R Code

↺ Start Over

▶ Run Code

```
1 # 1. Pooled: all collisions after, against all before
2 100 * (sum(sc$after) - sum(sc$before)) / sum(sc$before)
3
4 # 2. The average of each segment's own percentage change
5 #   (only segments that had a collision to begin with)
6 has_base <- sc$before > 0
7 mean(100 * (sc$after - sc$before)[has_base] / sc$before[has_base])
```

[1] -51.51149

[1] -35.55432

Predict before I run it: will the two answers be close?

Rule 3: check the denominator

- A percentage** is a claim about a base · find the base before you believe it
- Small bases** produce the largest percentages and the smallest changes
- Zero bases** produce no number at all · but the figure still has a bar there
- So ask** "how many segments is that percentage built on?"

The pooled change across all 200 package segments is -51.5%. The average of the 190 percentage changes that can be computed is -35.6%.

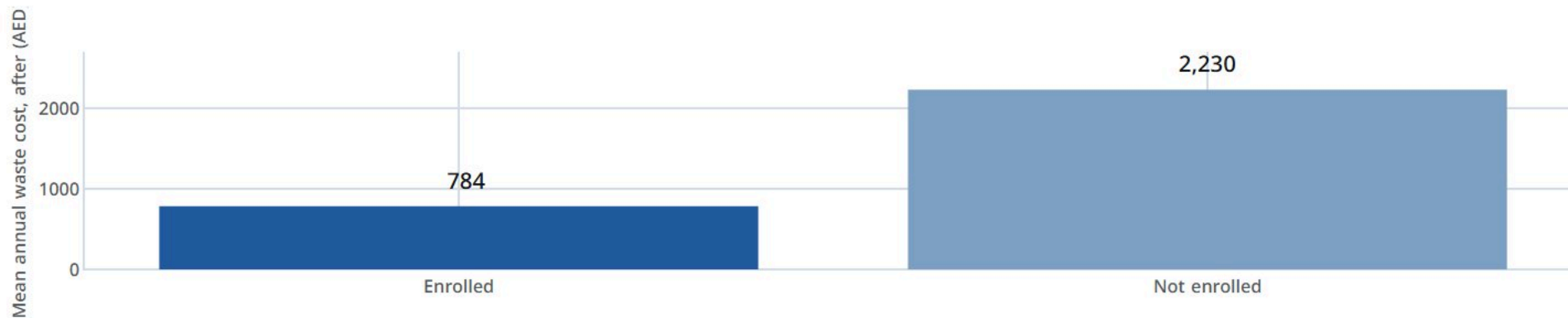
They differ by 16 percentage points because the pooled figure weights each segment by its collisions and the average gives every segment equal weight. **Which one belongs in the headline?**

4 • Levels or changes

The figure that flatters the programme

Day 1 Session 3's with-and-without comparison, drawn as a chart. After the programme, the enrolled businesses paid **784 AED** and the comparison businesses paid **2,230 AED**.

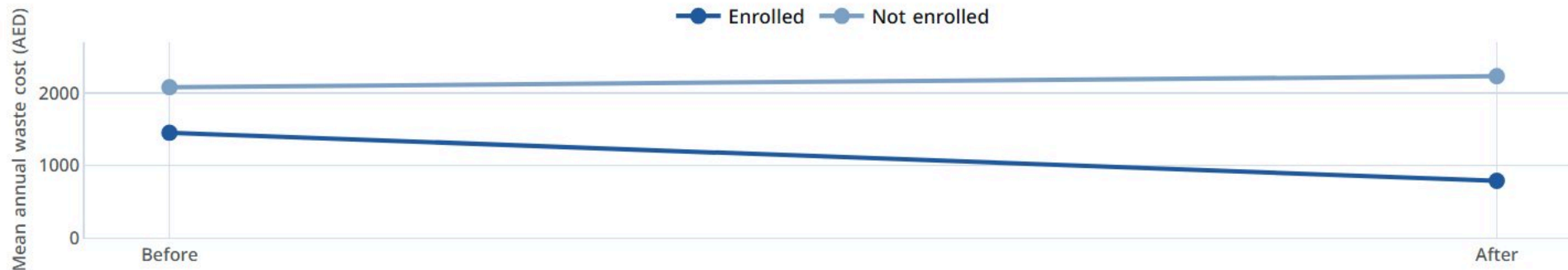
A difference of 1,447 AED. **The programme cut costs.**



Every number here is correct, and a figure like this could appear in a reputable report.

What question does it answer, and what question did the evaluation ask?

The same two groups, with where they started



The comparison group started 630 AED worse off, and its costs were rising anyway. The gap on the previous slide mixes the programme effect with that starting gap and with a trend unrelated to the programme. The DiD estimate from Day 2 Session 1 is 816 AED, not 1,447.

Day 1 showed this as a with-and-without comparison. **Why does the same comparison look more convincing as a bar chart?**

Rule 4: compare changes, not levels

Levels carry	whatever the two groups already differed by · and whatever moved on its own
Changes remove	the starting gap · if the groups were on parallel paths
Both are easy	to compute · only one answers "did the programme work?"
So ask	"what did the comparison group look like before?"

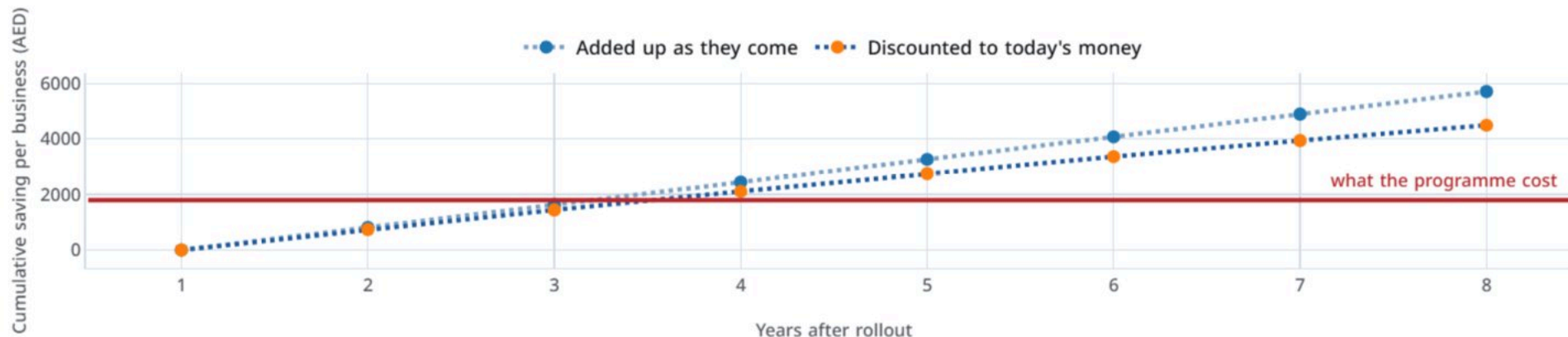
Check for four numbers on any before-and-after figure: before and after, for both groups.

Does the figure show the before for both groups? If not, what can it tell you about the programme's effect?

5 · The assumption the picture hides

Two ways to add up the same saving

From this morning: GreenWaste cost **1,800 AED** per business and saved **816 AED** a year, for five years starting one year after rollout. The two lines add up that saving in two ways.



Both lines use the same measured saving. The upper line counts a future saving as worth the same as one today and reaches 4,080 AED. The lower line discounts at 5% and reaches 3,365. The 715 AED gap is the **discount rate** at work.

Try it: change the discount rate



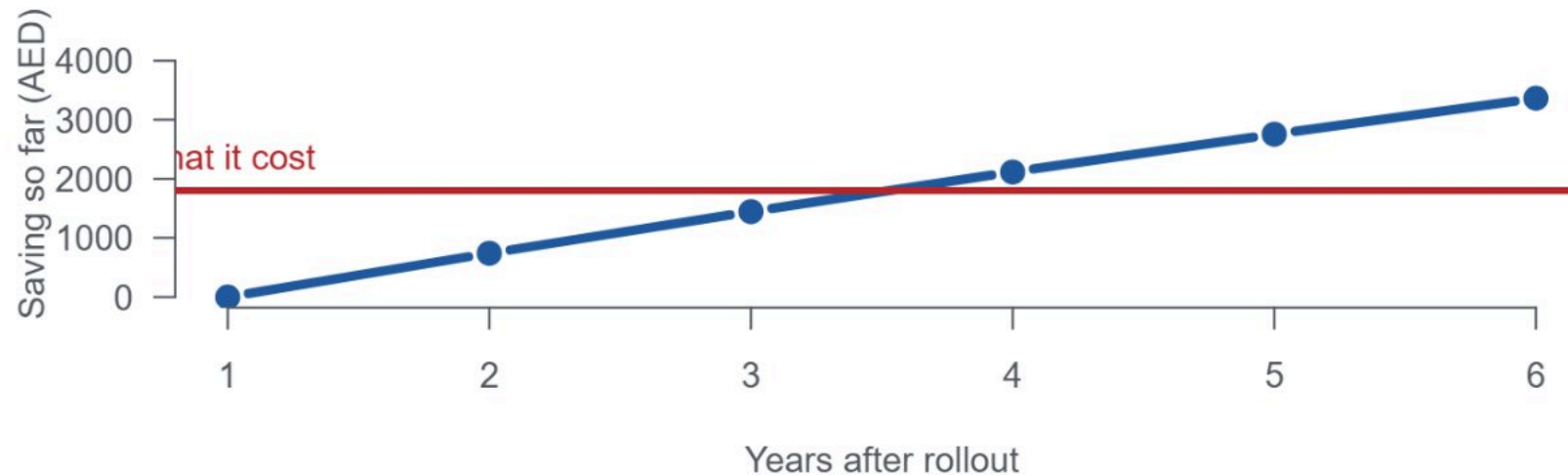
R Code

↺ Start Over

▶ Run Code

```
1 rate <- 0.05 # try 0, then 0.10
2 saving <- c(0, rep(816, 5)) # nothing in year 1, then five years
3 today <- saving / (1 + rate)^(1:6) # each year's saving in today's money
4 cat("Total in today's money:", round(sum(today)), "AED; cost 1,800 AED\n")
5 plot_payback(today)
```

Total in today's money: 3365 AED; cost 1,800 AED



Predict: what rate would push the total below the 1,800 AED cost?

Rule 5: discount in front of the reader

The ratio, discounted: $3,365 / 1,800 = 1.87$.

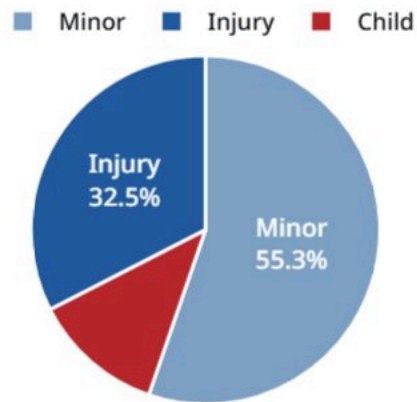
The undiscounted line is easier to explain and looks better. Leaving out the discount is a choice that favours the programme. **Would anyone in your review process ask the analyst to defend it, as they would a 5% rate?**

The cost line is crossed in year 3.2 on the undiscounted view and year 3.5 on the discounted one. **The discount rate alone moves payback by about four months.**

6 · The pie

What is this pie not telling you?

The school-zone evaluation reports its collisions by severity, as a pie chart.



In groups, three minutes. Write down two things this chart tells you and two things it does not.

What the pie cannot show

- It shows** which kind of collision is most common · minor ones
- It hides** that the categories overlap · a child collision is also an injury collision
- It hides** any change over time · both rounds pooled into one number
- It hides** the counts · a pie shows shares, not how many collisions

The first problem is the most serious. A pie chart says its slices add up to one whole. These do not: child collisions are also counted as injury collisions, so the slices overlap and the percentages are wrong.

The truncated axis and the level comparison were accurate charts that invited a wrong reading. This chart's percentages are wrong. **Before you read what a figure shows, how would you check that its parts add up?**

The same collisions, as counts



The same totals as bars. Child collisions sit **inside** the injury bar, because each one is also an injury collision, and the counts are on the chart.

In groups: what does this version show that the pie could not? What is still missing?

7 • Put AI to work

AI Snapshot: which chart is fairer?

Give the model figures A and B and this prompt: **“Which of these two charts better represents the difference between the two groups, and why?”**

A response like this is possible, especially with a short or leading prompt:

Both charts present the same underlying data accurately. **The truncated-axis chart is often preferable for a policy audience because it makes the difference visible, and a y-axis that starts at zero is not a requirement in professional data visualisation.**

On balance the truncated version communicates the finding more effectively, and a reader who wants the absolute values can consult the data labels.

Four minutes. Is it right? Does it answer the question that was asked? What does it explain well?

Debrief: what the answer skipped

Each sentence is accurate. The answer skips what decides the question:

- **who is reading it:** a minister and an analyst need different figures;
- **the sentence beside it:** a truncated axis suits "a difference exists", not "a large difference";
- **the number:** 630 AED, 1.13 SD, answers "is it large?" without any chart.

A better prompt names the reader and the sentence: *"This chart will sit beside the claim 'the programme reduced costs substantially'. Is the axis appropriate for that claim, for a non-technical reader?"*

Two habits: ask the AI to ask you questions before it answers, and paste its answer into a fresh chat to check it.

8 · The five rules

What to do with any figure you are handed

Rule	Why
1 Read the axis before the shape	A truncated axis makes a real gap look decisive. The gap is the same number either way; only your impression of it changes.
2 Ask what the bar is averaging over	A mean can sit between two groups that barely overlap. It summarises the group and may describe nobody in it.
3 Check the denominator	A percentage needs a base. Where the base is zero or tiny, the biggest percentage change belongs to the smallest number.
4 Compare changes, not levels	Levels carry whatever the two groups already differed by. The comparison the chart invites is often not the one that answers the question.
5 Discount the future in front of the reader	Undiscounted sums look bigger and are easier to draw. The gap between the two lines comes from the discount rate.

None of the five needs software, and you have now run all five on data from earlier in the week.

Which one would you find hardest to apply under time pressure?

The fastest check

The fastest of the five checks:

Read the y-axis, then read the sentence underneath the figure, then decide whether they are describing the same thing.

The axis says	how big the difference is on the page
The sentence says	how big the difference is in the world
They can disagree	when they do, the sentence overclaims
So ask	"large compared to what?"

Every figure this session had a fair version and a misleading version made from the same numbers. The difference came from choices about how to draw them.

Where this sits

- This morning** what a result is worth · cost, and the assumption that decides it
- Now** the form it arrives in · and what the form does to your reading
- Next session** the analyst is in the room · and has to answer for all of it
- Bring with you** the five rules, and the questions you did not get to ask

Pick one chart from a report your team received this year. Which of the five rules does it fail?

Next session the 3ie trainer plays the evaluator of a study you have not seen. Which questions from your checklist, and from today, will you ask first?